EE4-07 Coding Theory

Instructions for Candidates

Answer all five questions. Each question carries 20 marks.

The star notation * right after the sub-question numbering means that the particular sub-question may be difficult to solve.

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1. (Finite Fields)

   (a) Let $f(x) = x^3 + x^2 + 2 \in \mathbb{F}_3[x]$ and $g(x) = x^2 + 2 \in \mathbb{F}_3[x]$.

      i Find the greatest common divisor $h(x)$ of $f(x)$ and $g(x)$, i.e., $h(x) = \gcd(f(x), g(x))$. Write $h(x)$ as a *monic* polynomial. [4]

      ii Find the polynomials $a(x) \in \mathbb{F}_3[x]$ and $b(x) \in \mathbb{F}_3[x]$ such that $h(x) = a(x) f(x) + b(x) g(x)$. [4]

   (b) Use Bézout's identity to prove Euclid's Lemma:

   Let $r_1, r_2 \in \mathbb{Z}^+$ and $\gcd(r_1, r_2) = 1$. If $r_1 | (r_2 r)$, then $r_1 | r$. [2]

   (c) Let $f(x) = x^2 + 1 \in \mathbb{F}_2[x]$. Is this polynomial irreducible? Justify your answer. [2]

   (d) Let $f(x) = x^2 + 1 \in \mathbb{F}_3[x]$. Is this polynomial irreducible? Justify your answer. [2]

   (e) Consider the polynomial ring $\mathcal{R} := \mathbb{F}_p[x]/f(x)$ where $f(x) \in \mathbb{F}_p[x]$ has degree larger than one.

      i Prove that if $f(x) \in \mathbb{F}_p[x]$ is irreducible, then $\mathcal{R}$ is a field. [3]

      ii Prove that if $\mathcal{R}$ is a field, then $f(x) \in \mathbb{F}_p[x]$ is irreducible. [3]

2. (Cryptography)

(a) Let p be a prime number. For given $b, y \in \mathbb{F}_p^*$, define the discrete logarithmic function $x = \log_b y \bmod p$ if $b^x = y \bmod p$.

i Let $p = 7$ and $\alpha = 3 \in \mathbb{F}_p$. Find $\text{ord}(\alpha)$ by computing α^x, $x = 1, 2, \cdots$. [2]

ii Let $p = 7$ and $b = 3$. Compute $\log_b y \bmod p$ for $y = 1, 2, 3$ respectively. [2]

iii Let $p = 7$ and $\alpha = 2 \in \mathbb{F}_p$. Find $\text{ord}(\alpha)$ by computing α^x, $x = 1, 2, \cdots$.. [2]

iv Let $p = 7$ and $b = 2$. Compute $\log_b y \bmod p$ for $y = 1, 2, 3$ respectively. [2]

v Prove that if b is a primitive element, then $b^{x_1} \neq b^{x_2}$ for all $0 \leq x_1 < x_2 \leq p - 1$. [2]

vi Explain how to choose the base b for the discrete logarithm function so that it is well defined. [2]

(b) Suppose that Alice would like to save her password securely on a server. Denote her user name by i and the raw password by x_i. What information should be stored on the server? [2]

(c) Consider Shamir's Secret Sharing scheme to share a secret $S \in \mathbb{F}_p^*$ among n users:

Randomly choose $k - 1$ integers $a_1, \cdots, a_{k-1} \in \mathbb{F}_p^*$. Set $a_0 = S$. Set $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_{k-1} x^{k-1}$. Evaluate $f(x)$ at n distinct points to obtain $(t_i, f(t_i))$, $t_i \in \mathbb{F}_p^*$ and $i = 1, \cdots, n$.

i How many pairs $(t_i, f(t_i))$ are needed in order to uniquely recover the secret S? [2]

ii Given ℓ pairs $(t_{i_1}, f(t_{i_1})), (t_{i_2}, f(t_{i_2})), \cdots, (t_{i_\ell}, f(t_{i_\ell}))$, a linear system $\boldsymbol{aM} = \boldsymbol{f}$ can be used to find the polynomial coefficients, where $\boldsymbol{a} = [a_0, \cdots, a_{k-1}]$ and $\boldsymbol{f} = [f(t_{i_1}), f(t_{i_2}), \cdots, f(t_{i_\ell})]$. Write the explicit form of the matrix $\boldsymbol{M}$. [2]

iii Use your result for the Problem 2.(c)-ii to justify your answer to Problem 2.(c)-i. You are allowed to use the properties of Vandermonde matrix. [2]

3. (Linear Codes)

(a) Let $\mathcal{C} \subset \mathbb{F}_q^n$ be a linear code with distance d. State the relationship between d and the weights of the codewords in the code. (No proof is needed.) [2]

(b) Let $\mathcal{C} \subset \mathbb{F}_q^n$ be a linear code with distance d. Let $\boldsymbol{H}$ be its parity-check matrix. State the relationship between d and the linear dependence (or independence) of the columns of $\boldsymbol{H}$. (No proof is needed.) [2]

(c) Let $\mathcal{C} \subset \mathbb{F}_2^7$ be a linear code generated by the matrix

$$\boldsymbol{G} = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

i Use Gaussian elimination to change the generator matrix into the form of $\boldsymbol{G}' = [\boldsymbol{A}\ \boldsymbol{I}]$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding parity-check matrix $\boldsymbol{H}$ in the systematic form. [2]

iii Assume that a message $\boldsymbol{m}_1$ is encoded into a codeword $\boldsymbol{c}_1$ using $\boldsymbol{G}'$. Let the received word be $\boldsymbol{y}_1 = [1\ 1\ 0\ 1\ 1\ 0\ 1]$. Compute the syndrome vector $\boldsymbol{s}_1$. Find the output of the minimum (Hamming) distance decoding, say $\hat{\boldsymbol{c}}_1$, and the corresponding transmitted message $\hat{\boldsymbol{m}}_1$. [3]

iv Assume that a message $\boldsymbol{m}_2$ is encoded into a codeword $\boldsymbol{c}_2$ using $\boldsymbol{G}'$. The codeword $\boldsymbol{c}_2$ is transmitted over an erasure channel and the received word is given by $\boldsymbol{y}_2 = [1\ ?\ 0\ ?\ 0\ 0\ 1]$. Set the question marks in $\boldsymbol{y}_2$ to zero and compute the corresponding syndrome vector $\boldsymbol{s}_2$. Find the transmitted codeword $\boldsymbol{c}_2$ and the message $\boldsymbol{m}_2$. [3]

(d) * Define

$$\mathcal{C}_2 = \left\{ \left(c_1, \cdots, c_n, \sum_{i=1}^{n} c_i \right) : (c_1, \cdots, c_n) \in \mathcal{C} \right\},$$

where $\mathcal{C}$ is the code defined in Problem 3.(c).

i Find the length of the codewords in $\mathcal{C}_2$, denoted by n_2. [1]

ii Find the dimension of $\mathcal{C}_2$ defined as $k_2 := \log_2 |\mathcal{C}_2|$ where $|\mathcal{C}_2|$ gives the number of codewords in $\mathcal{C}_2$. [1]

iii Find the generator matrix $\boldsymbol{G}_2$ of $\mathcal{C}_2$ using the $\boldsymbol{G}$ from Problem (c). (No proof is needed.) [2]

iv Find the distance of $\mathcal{C}_2$. Prove your answer using the result for Problem 3.(a). [2]

4. (RS, Cyclic, and BCH Codes)

(a) Consider a linear code with parameters $[n, k, d]$. The Singleton bound states that $d \leq n - k + 1$. Prove it. [3]

(b) A Reed-Solomon code can be defined as follows. Let $\mathbb{F}_q$ be a finite field and α be a primitive element. Let $n = q - 1$. For a given polynomial $f(x) \in \mathbb{F}_q[x]$, define the evaluation mapping $\text{eval}(f)$ by

$$\begin{aligned} &\mathbb{F}_q[x] \to \mathbb{F}_q^n \\ &\quad f \mapsto \boldsymbol{c} = [c_0, c_1, \cdots, c_{n-1}], \text{ where } c_i = f\left(\alpha^i\right). \end{aligned}$$

An $[n, k]$ Reed-Solomon code is defined as $\mathcal{C} = \{\text{eval}(f),\ 0 \leq \deg(f) \leq k - 1\}$.

i Prove that Reed-Solomon codes are linear codes. [3]

ii Prove that Reed-Solomon codes achieve the Singleton bound. [3]

(c) Let $q = 3$ and $n = 26$. Construct a BCH code in the following way.

i Write down the cyclotomic cosets $C_0, C_1, \cdots, C_8$ of 3 modulo 26. [4]

ii Let α be a primitive element of $\mathbb{F}_{27}$. Define $M^{(i)}(x) = \prod_{j \in C_i}(x - \alpha^j)$. Let $g(x) = \text{lcm}\left(M^{(1)}(x), \cdots, M^{(8)}(x)\right)$. Consider the cyclic code $\mathcal{C}$ generated by $g(x)$.

A. Find the degree of $g(x)$. [2]

B. Decide the dimension k of the generated code $\mathcal{C}$. [2]

C. Find the tightest lower bound on the distance d of the code $\mathcal{C}$. Prove your result. You are allowed to use the properties of Vandermonde matrix. [3]

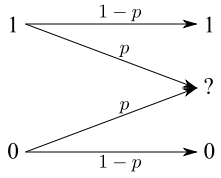
5. (Channel Polarization)

Recall the definition

$$
\begin{aligned}
H(X) &:= -\sum_{x\in\mathcal{X}} p_X(x)\log_2 P_X(x),\\
H(X|y) &:= -\sum_{x\in\mathcal{X}} p_{X|Y}(x|y)\log_2 p_{X|Y}(x|y),\\
H(X|Y) &:= -\sum_{y\in\mathcal{Y}} p_Y(y)\,H(X|y).\\
I(X;Y) &:= H(Y) - H(Y|X) = H(X) - H(X|Y).
\end{aligned}
$$

Define $H(p) := -p\log_2 p - (1-p)\log_2(1-p)$. Note that $0\log_2 0 = 0$.

(a) Consider the BEC channel:



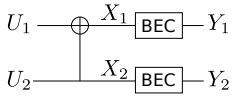
Assume that $p_X(0) = p_X(1) = \frac{1}{2}$.

i Find $p_Y(y)$ for $y \in \{0, 1, ?\}$. [3]

ii Find the cases that $H(X|y) = 0$ and $H(X|y) = 1$ respectively. [2]

iii Find $I(X;Y)$. [2]

(b) * Consider the following channel of which the input $u_1u_2 \in \{0,1\}^2$ and the output $y_1y_2 \in \{0,1,?\}^2$:



Assume that U_1, U_2 are independent with distribution $p_U(0) = p_U(1) = \frac{1}{2}$.

i Find $p_{Y_1Y_2}(y_1y_2)$ when y_1y_2 varies in $\{0,1,?\}^2$. [3]

ii It is straightforward to see that $H(U_1|y_1y_2)$ can only take two values 0 and 1. Find the cases that $H(U_1|y_1y_2) = 0$ and $H(U_1|y_1y_2) = 1$ respectively. [3]

iii Find $I(U_1; Y_1Y_2)$. [2]

iv It is straightforward to see that $H(U_2|y_1y_2u_1)$ can only take two values 0 and 1. Find the cases that $H(U_2|y_1y_2u_1) = 0$ and $H(U_2|y_1y_2u_1) = 1$ respectively. [3]

v Find $I(U_2; Y_1 Y_2 U_1)$. [2]

Solution of Question 1.

(a)

i It is straightforward to compute that

$$x^3 + x^2 + 2 = (x+1)\left(x^2+2\right) + x,$$
$$x^2 + 2 = x \cdot x + 2.$$

As a result,

$$1 = \gcd\left(f(x), g(x)\right).$$

[4]

Remark. Some students forgot to present their final results in monic polynomials.

ii According to the previous part, it is clear that

$$\begin{aligned} 2 &= x^2 + 2 + 2x \cdot x \\ &= x^2 + 2 + 2x\left(x^3 + x^2 + 2 + (2x+2)\left(x^2+2\right)\right) \\ &= 2x\left(x^3 + x^2 + 2\right) + \left(x^2 + x + 1\right)\left(x^2 + 2\right). \end{aligned}$$

Multiply both sides with 2. It holds that

$$1 = x\left(x^3 + x^2 + 2\right) + \left(2x^2 + 2x + 2\right)\left(x^2 + 2\right).$$

As a result,

$$a(x) = x,$$
$$b(x) = 2x^2 + 2x + 2.$$

[4]

(b) By Bézout's identity, there exist $a, b \in \mathbb{Z}$ such that $1 = ar_1 + br_2$. Multiply both sides by r. One obtains that $r = ar_1 r + br_2 r$. By assumption $r_1 \mid (r_2 r)$, it is clear that r_1 divides the right hand side of the equation. Hence $r_1 | r$. [2]

(c) Since $f(x) = (x+1)(x+1)$, $f(x)$ is reducible. [2]

(d) $f(x) = x^2 + 1 \in \mathbb{F}_3[x]$ is irreducible. To justify it, one can try all possible polynomials $g(x)$ in $\mathbb{F}_3[x]$ of degree $0 < \deg(g) < 2$. As $f(x) = x \cdot x + 1$

and $f(x) = (x+1)(x+2) + 2$, we conclude that $f(x)$ is irreducible. [2]

Remark. To show that $f(x)$ is reducible, it is fine to give just one example such that $f(x) = g(x) \cdot h(x)$. However, it is much more difficult to show a polynomial is irreducible as it requires exhaustive search in general.

(e)

i Since f is irreducible, it holds that $\gcd(f, g) = 1$ for all $g \in \mathcal{R}$. By Bézout's identity, it holds that

$$1 = c(x) f(x) + d(x) g(x),$$

or $d(x) g(x) = 1 \bmod f(x)$. This proves the existence of g^{-1}. Hence $\mathcal{R}$ is a field. [3]

ii Suppose that $f(x)$ is reducible, i.e., there exist $g(x)$ and $h(x)$ such that $f(x) = g(x) h(x)$ and $1 \leq \deg(g(x)) < \deg(f(x))$ and $1 \leq \deg(h(x)) < \deg(f(x))$. Since $\mathcal{R}$ is a field, $g^{-1}(x)$ exits. Hence $h(x) = (g^{-1}(x) g(x)) h(x) = g^{-1}(x) f(x) = g^{-1}(x) \cdot 0 = 0 \bmod f(x)$. This contradicts that $1 \leq \deg(h(x))$. Hence $f(x)$ must be irreducible. [3]

Remark. This problem is testing the students basic knowledge in Euclidean algorithm and finite fields.

Solutions of Question 2.

(a)

i Since $3^1 = 3$, $3^2 = 2$, $3^3 = 6$, $3^4 = 4$, $3^5 = 5$, $3^6 = 1$, one has $\operatorname{ord}(3) = 6$. [2]

ii $\log_3 1 = 6$, $\log_3 2 = 2$, and $\log_3 3 = 1$. [2]

iii Since $2^1 = 2$, $2^2 = 4$, $2^3 = 1$, one has $\operatorname{ord}(2) = 3$. [2]

iv $\log_2 1 = 3$, $\log_2 2 = 1$, and $\log_2 3$ does not exist. [2]

v Suppose that there exist x_1, x_2 such that $0 \leq x_1 < x_2 \leq p-1$ and $b^{x_1} = b^{x_2}$. Then $b^{x_2}/b^{x_1} = b^{x_2 - x_1} = 1$ which suggests that $\operatorname{ord}(b) \leq x_2 - x_1 < p-1$. This contradicts the definition of a primitive element b. The claim is therefore proved. [2]

vi The base b should be a primitive element. Suppose that b is not a primitive element. Then the set $\mathcal{B} = \{b^1, b^2, \cdots\}$ contains only $\operatorname{ord}(b)$ many elements with $\operatorname{ord}(b) < p-1$. This means that for some element $y \in \mathbb{F}_p^*$, $\log_b y$ is not well defined. [2]

Remark. This part of the problem is designed to show the "randomness" of the discrete logarithmic function.

(b) Alice would like to store (i, b^{x_i}) on the server. [2]

(c)

i At least k pairs are needed. [2]

ii The linear system is given by

$$[a_0, a_1, \cdots, a_{k-1}] \begin{bmatrix} 1 & 1 & \cdots & 1 \\ t_{i_1} & t_{i_2} & \cdots & t_{i_\ell} \\ \vdots & \vdots & \ddots & \vdots \\ t_{i_1}^{k-1} & t_{i_2}^{k-1} & \cdots & t_{i_\ell}^{k-1} \end{bmatrix} = \left[f(t_{i_1}), f(t_{i_2}), \cdots, f(t_{i_\ell})\right].$$

[2]

iii When $\ell = k$, the matrix $\boldsymbol{M}$ is a Vandermonde matrix and hence its inverse exists. The vector $\boldsymbol{a}$ can be obtained from $\boldsymbol{a} = \boldsymbol{f}\boldsymbol{M}^{-1}$. When $\ell < k$, the number of equations is less than the number of unknowns. The solution of this linear system is not unique. [2]

Remark. This part of the problem uses Vandermonde matrix argument to prove that Shamir's Secret Sharing scheme works fine. It has been covered in lectures.

Solutions of Question 3.

(a)

$$d(\mathcal{C}) = \min_{\boldsymbol{c}\in\mathcal{C},\, \boldsymbol{c}\neq\mathbf{0}} \text{weight}(\boldsymbol{c}).$$

[2]

Remark. Many students forgot the condition $\boldsymbol{c} \neq \mathbf{0}$.

(b) $d(\mathcal{C})$ is the minimum number of linearly dependent columns of $\boldsymbol{H}$, i.e., $d = \text{spark}(\boldsymbol{H})$. [2]

Remark. Some students forgot "minimum". Also, it is not necessarily related to maximum number of linear independent columns. This has been discussed in the lectures. To highlight the point, note that in the following example

$$\boldsymbol{H} = \begin{bmatrix} & & 1 \\ & \boldsymbol{I} & 0 \\ & & \vdots \\ & & 0 \end{bmatrix} \in \mathbb{F}_2^{n\times(n+1)},$$

the rank (maximum number of linearly independent columns) is n while the spark is two.

(c)

i By Gaussian elimination, it is clear that

$$\boldsymbol{G}' = \begin{bmatrix} 1 & 1 & 1 & 1 & & & \\ 1 & 1 & 0 & & 1 & & \\ 1 & 0 & 1 & & & 1 & \\ 0 & 1 & 1 & & & & 1 \end{bmatrix}.$$

[2]

ii

$$\boldsymbol{H} = \begin{bmatrix} 1 & & & 1 & 1 & 1 & 0 \\ & 1 & & 1 & 1 & 0 & 1 \\ & & 1 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

[2]

iii The syndrome vector is given by

$$\boldsymbol{s}_1 = \boldsymbol{y}_1\boldsymbol{H}^T = [1\ 0\ 0].$$

As a result, the error vector is given by $\boldsymbol{e} = [1\ 0\ 0\ 0\ 0\ 0\ 0]$ and the minimum distance decoder outputs $\hat{\boldsymbol{c}}_1 = [0\ 1\ 0\ 1\ 1\ 0\ 1]$. From the last four bits of $\hat{\boldsymbol{c}}_1$, it is clear that $\hat{\boldsymbol{m}}_1 = [1\ 1\ 0\ 1]$. [3]

iv The syndrome vector is given by

$$[1\ 0\ 0\ 0\ 0\ 0\ 1]\,\boldsymbol{H}^T = [1\ 1\ 1]\,.$$

Let c_2 and c_4 be the 2nd and 4th symbols in $\boldsymbol{c}$. Then one has

$$[c_2\ c_4]\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = [1\ 1\ 1]\,.$$

It is clear that $[c_2\ c_4] = [0\ 1]$. Hence $\boldsymbol{c} = [1\ 0\ 0\ 1\ 0\ 0\ 1]$ and $\boldsymbol{m} = [1\ 0\ 0\ 1]$. [3]

(d)

i $n_2 = 8$. [1]

ii $k_2 = 4$. The dimension of $\mathcal{C}_2$ is the same as that of $\mathcal{C}$ because there is one-to-one mapping between the codewords in $\mathcal{C}$ and those in $\mathcal{C}_2$. [1]

iii

$$\boldsymbol{G}_2 = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

[2]

iv $d(\mathcal{C}_2) = 4$. For any $\boldsymbol{c} \in \mathcal{C}$, let $\boldsymbol{c}_2 \in \mathcal{C}_2$ be the corresponding codeword of which the first seven bits are given by $\boldsymbol{c}$. That $d(\mathcal{C}_2) = 4$ can be obtained by observing that

- $\boldsymbol{c} = \boldsymbol{0} \Rightarrow \boldsymbol{c}_2 = \boldsymbol{0}$.
- $\mathrm{wt}(\boldsymbol{c}) = 3 \Rightarrow \mathrm{wt}(\boldsymbol{c}_2) = 4$.
- $\mathrm{wt}(\boldsymbol{c}) \geq 3$, for all $\boldsymbol{c} \in \mathcal{C}$ and $\boldsymbol{c} \neq \boldsymbol{0} \Rightarrow \mathrm{wt}(\boldsymbol{c}_2) \geq 4$ for all $\boldsymbol{c}_2 \in \mathcal{C}_2$ and $\boldsymbol{c}_2 \neq \boldsymbol{0}$. [2]

Remark. Most students had difficulty to give the correct answers. This part of the problem is perhaps the most difficult part in the whole paper.

Solutions of Question 4.

(a) The parity-check matrix $\boldsymbol{H} \in \mathbb{F}^{(n-k)\times n}$ contains $n-k$ rows. Every $n-k+1$ columns must be linearly dependent. Hence $d \leq n-k+1$. [3]

(b)

i Let $\boldsymbol{c}_1 = \text{eval}(f_1)$ and $\boldsymbol{c}_2 = \text{eval}(f_2)$ where $\deg(f_1) \leq k-1$ and $\deg(f_2) \leq k-1$. Then $\alpha\boldsymbol{c}_1 + \beta\boldsymbol{c}_2 = \text{eval}(g)$ with $g = \alpha f_1 + \beta f_2$. Since $\deg(g) \leq k-1$, $\text{eval}(g) \in \mathcal{C}$. [3]

ii A polynomial of degree $k-1$ can have at most $k-1$ zeros. Hence, $\forall \boldsymbol{c} \in \mathcal{C}$ s.t. $\boldsymbol{c} \neq 0$, $\boldsymbol{c} = \text{eval}(f)$ has weight at least $n-k+1$. [3]

(c)

i $C_0 = \{0\}$, $C_1 = C_3 = \{1,3,9\}$, $C_2 = C_6 = \{2,6,18\}$, $C_4 = \{4,12,10\}$, $C_5 = \{5,15,19\}$, $C_7 = \{7,21,11\}$, $C_8 = \{8,24,20\}$. [4]

ii

A. $\deg(g) = |C_1| + |C_2| + |C_4| + |C_5| + |C_7| + |C_8| = 18$. [2]

B. $k = n - \deg(g) = 26 - 18 = 8$. [2]

C. $d \geq 13$. Note that $\alpha^1, \alpha^2, \cdots, \alpha^{12}$ are roots of $g(x)$. For any code $\boldsymbol{c} \in \mathcal{C}$, all the roots of $g(x)$ are the roots of the corresponding generating function $c(x)$. One has

$$\underbrace{\begin{bmatrix} 1 & \alpha & \cdots & \alpha^{(n-1)} \\ 1 & \alpha^2 & \cdots & \alpha^{2(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{12} & \cdots & \alpha^{12(n-1)} \end{bmatrix}}_{\boldsymbol{A}} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{bmatrix} = \mathbf{0}.$$

Since every 12-column sub-matrix of $\boldsymbol{A}$ is a Vandermonde matrix and has full column rank, $d(\mathcal{C}) \geq 12 + 1 = 13$. [3]

Remark. The motivation to introduce the BCH codes in the lectures is the lower bound on the distance of BCH codes. Unfortunately, many students had difficulties to answer this question.

Solutions of Question 5.

(a)

i $p_Y(0) = p_Y(1) = \frac{1}{2}(1-p)$ and $p_Y(?) = p$. [3]

ii It is clear that $H(X|0) = H(X|1) = 0$ and $H(X|?) = 1$. [2]

iii Since $H(X|Y) = p$, $I(X;Y) = 1-p$. [2]

(b)

i $p_{Y_1Y_2}(00) = p_{Y_1Y_2}(01) = p_{Y_1Y_2}(10) = p_{Y_1Y_2}(11) = \frac{1}{4}(1-p)^2$.
$p_{Y_1Y_2}(?0) = p_{Y_1Y_2}(?1) = \frac{1}{2}p(1-p)$.
$p_{Y_1Y_2}(0?) = p_{Y_1Y_2}(1?) = \frac{1}{2}p(1-p)$.
$p_{Y_1Y_2}(??) = p^2$. [3]

ii If neither y_1 nor y_2 is ?, then we can identify u_2 and u_1 uniquely. Hence $H(U_1|00) = H(U_1|01) = H(U_1|10) = H(U_1|11) = 0$.
If either y_1 or y_2 is ?, U_1 has equal probability to be 0 or 1. Hence $H(U_1|?0) = H(U_1|?1) = H(U_1|0?) = H(U_1|1?) = H(U_1|??) = 1$. [3]

iii Since $H(U_1|Y_1Y_2) = 2p - p^2$, $I(U_1;Y_1Y_2) = 1 - 2p + p^2$. [2]

iv If $y_2 \neq ?$, then u_2 is uniquely identified. Hence $H(U_2|y_1y_2u_1) = 0$ if $y_2 \neq ?$.
If $y_2 = ?$ but $y_1 \neq ?$, we are still able to find u_2 via $y_1 = u_1 + u_2$. Hence $H(U_2|y_1?u_1) = 0$ when $y_1u_1 \in \{0,1\}^2$.
If $y_1 = y_2 = ?$, $H(U_2|??u_1) = 1$. [3]

v Since $H(U_2|Y_1Y_2U_1) = p^2$, $I(U_2;Y_1Y_2U_1) = 1 - p^2$. [2]

Remark. This problem is designed to test the students the understanding of the new materials that were introduced in this academic year. To answer this problem correctly, one needs to clearly understand the difference between conditional probability and joint probability (They are linked by Bayes' rule.).